



Vectors in Kinematics · Mark Scheme

Mechanics Chapter 8: Vectors in Kinematics. M = method mark A = accuracy mark (depends on M) B = independent mark ft = follow through

Section A – Position, velocity and acceleration

- 1 Speed $\sqrt{9+16} = 5 \text{ ms}^{-1}$ B1; 53.1° below **i** B1.
- 2 Displacement $6\mathbf{i} - 8\mathbf{j}$ M1; $\mathbf{v} = (3\mathbf{i} - 4\mathbf{j}) \text{ ms}^{-1}$ A1; speed 5 ms^{-1} A1.
- 3 $\mathbf{v} = 6t\mathbf{i} + 5\mathbf{j}$ M1 A1; at $t = 2$: $\sqrt{144+25} = 13 \text{ ms}^{-1}$ A1.
- 4 (a) $\mathbf{r} = (2+4t)\mathbf{i} + (3-2t)\mathbf{j}$ B1. (b) $3-2t = 0 \Rightarrow t = 1.5 \text{ h}$ M1; position $8\mathbf{i}$ km A1. (c) Bearing $90^\circ + \tan^{-1} \frac{2}{4} = 117^\circ$ (116.6°) B1.
- 5(a) $\mathbf{r}_B - \mathbf{r}_A$ M1; $= (8-4t)\mathbf{i} + (1-t)\mathbf{j}$ A1.
- 5(b) i-component zero: $8-4t = 0$ M1; $t = 2$ A1.
- 5(c) $|1-2| = 1 \text{ km}$ B1.
- 5(d) $d^2 = (8-4t)^2 + (1-t)^2$ M1; $= 64 - 64t + 16t^2 + 1 - 2t + t^2 = 17t^2 - 66t + 65$ A1.

Section B – Calculus with vectors

- 6 $\mathbf{v} = 3t^2\mathbf{i} + 4t\mathbf{j}$ B1; $\mathbf{a} = 6t\mathbf{i} + 4\mathbf{j}$ B1.
- 7 $\mathbf{a} = 2\mathbf{i} - 2t\mathbf{j}$ M1; at $t = 1$: $2\mathbf{i} - 2\mathbf{j}$ A1; magnitude $2\sqrt{2} = 2.83 \text{ ms}^{-2}$ A1.
- 8 $\mathbf{v} = 3t^2\mathbf{i} - 2t\mathbf{j} + \mathbf{c}$ M1; $\mathbf{c} = \mathbf{i} + \mathbf{j}$ A1; at $t = 2$: $13\mathbf{i} - 3\mathbf{j} \text{ ms}^{-1}$ A1.
- 9 (a) $\mathbf{v} = (3\mathbf{i} + 4\mathbf{j}) + 4(2\mathbf{i} - \mathbf{j}) = 11\mathbf{i}$ M1 A1; no **j**-component, so parallel to **i** A1. (b) $\mathbf{r} = 4(3\mathbf{i} + 4\mathbf{j}) + 8(2\mathbf{i} - \mathbf{j}) = 28\mathbf{i} + 8\mathbf{j}$ B1.
- 10(a) $\mathbf{a} = 6t\mathbf{i} + 2\mathbf{j}$; at $t = 1$: $6\mathbf{i} + 2\mathbf{j}$ B1; magnitude $\sqrt{40} = 6.32 \text{ ms}^{-2}$ B1.
- 10(b) i-component zero: $3t^2 = 12$ M1; $t = 2$ A1.
- 10(c) $\mathbf{r} = (t^3 - 12t)\mathbf{i} + (t^2 - 5t)\mathbf{j} + \mathbf{c}$ M1; $\mathbf{c} = \mathbf{i} + 2\mathbf{j}$ A1; at $t = 2$: $-15\mathbf{i} - 4\mathbf{j}$ A1.
- 10(d) $\mathbf{v} = 15\mathbf{i} + \mathbf{j}$ M1; speed $\sqrt{226} = 15.0 \text{ ms}^{-1}$ A1.

Section C – Projectiles using vectors

- 11 $\mathbf{v} = (10\mathbf{i} + 5\mathbf{j}) - 9.8\mathbf{j}$ M1; $= 10\mathbf{i} - 4.8\mathbf{j} \text{ ms}^{-1}$ A1.
- 12 $\mathbf{r} = 10t\mathbf{i} + (5t - 4.9t^2)\mathbf{j}$ M1 A1; at $t = 2$: $20\mathbf{i} - 9.6\mathbf{j}$ A1.
- 13 **j**-component: $14.7t - 4.9t^2 = 0$ M1; $t = 3 \text{ s}$ A1; distance 36 m A1.
- 14 Vertical velocity zero: $20 - 9.8t = 0$, $t = 2.04$ M1 A1; greatest height $\frac{400}{19.6} = 20.4 \text{ m}$ A1; $\mathbf{r} = 30.6\mathbf{i} + 20.4\mathbf{j}$ A1.
- 15(a) $14t = 28 \Rightarrow t = 2$ M1; $h = 21(2) - 4.9(4)$ M1; $h = 22.4$ A1.
- 15(b) $\mathbf{v} = 14\mathbf{i} + (21 - 19.6)\mathbf{j}$ M1; $= 14\mathbf{i} + 1.4\mathbf{j}$ A1.
- 15(c) $\tan^{-1} \frac{1.4}{14} = 5.71^\circ$ B1; rising, since the **j**-component is positive B1.
- 15(d) $\sqrt{14^2 + 1.4^2} = 14.1 \text{ ms}^{-1}$ B1.
- 15(e) $\frac{21^2}{19.6} = 22.5 \text{ m}$ B1.

Roar-level challenge

- 16 $\mathbf{r}_A = 2t\mathbf{i} + 3t\mathbf{j}$, $\mathbf{r}_B = (10-3t)\mathbf{i} + (-4+5t)\mathbf{j}$ M1 A1; **i**: $2t = 10-3t \Rightarrow t = 2$ M1; **j** at $t = 2$: $6 = -4 + 10$, so they collide A1; at $4\mathbf{i} + 6\mathbf{j}$ A1.